

NAME OF THE PROGRAMME	B.E/B.TECH .COMPUTER SCIENCE AND ENGINEERING/INFORMATION TECHNOLOGY/CYBER SECURITY/COMPUTER SCIENCE AND BUSINESS SYSTEMS
YEAR	II
SEMESTER	IV
REGULATIONS	2024
COURSE CODE	CS4402/
COURSE NAME	ADVANCE JAVA PROGRAMMING

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REVISED BLOOMS TAXONOMY(RBT)

L1- Remembering	L2 - Understanding	L3 – Applying	L4 – Analyzing	L5 – Evaluating	L6 – Creating
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Course Objectives:

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1.	To enable students to understand and apply AWT components, layouts, graphics, and event-handling mechanisms to create basic GUI applications
2.	To help students understand and utilize Swing components and containers for building interactive GUI applications.
3.	To enable students to understand and apply servlet lifecycle, request–response handling, and database connectivity in web applications
4.	To provide students the ability to understand and apply JSP architecture, scripting elements and JSTL for building dynamic web pages.
5.	To develop the ability to understand and apply modern Java frameworks such as Spring, Hibernate, and Maven for enterprise application development.

Course Outcomes:

On Completion of the course the students will be able to

CO. NO	Course Outcomes	RBT Level
1.	Design and develop simple event-driven GUI applications using AWT and Java's delegation event model.	L5
2.	Construct user-friendly desktop applications using Swing's advanced components and layout mechanisms..	L5
3.	Develop dynamic server-side applications using servlets with session management and database interaction.	L6
4.	Build dynamic and data-driven web interfaces using JSP and JSTL tags.	L6
5.	Develop modular and scalable applications using Spring MVC/Spring Boot, Hibernate ORM, and Maven build tools	L6

UNIT	TITLE
III	Jave Servlets

S N O	QUESTIONS	RB T	CO	MARK	TOPIC	IMA GES
1.	State purpose of servlet in java	L2	CO3	2	Java EE Architecture	
2.	Compare Enterprise Application and Enterprise Edition	L2	CO3	2	Java EE Architecture	
3.	List common tasks performed by Servlet Container?	L2	CO3	2	Servers and Containers	
4.	Describe web application and what is its directory structure?	L2	CO3	2	Java EE Architecture	
5.	Differentiate ServletConfig and ServletContext?	L1	CO3	2	Servlet API	
6.	Differentiate GET and POST method?	L2	CO3	2	Servlet API	
7.	List life cycle methods of a servlet?	L1	CO3	2	Servlet API and Lifecycle	
8.	State the role of Servlets in the Model-View-Controller (MVC) design pattern	L1	CO3	2	Java EE Architecture	
9.	How to get the IP address of client in servlet?	L3	CO3	2	Handling Client Requests and Responses	
10.	What is Request Dispatcher?	L1	CO3	2	Request Dispatcher	

	Compare PrintWriter and ServletOutputStream	L2	CO3	2	Handling Client Requests and Responses	
12.	List use of servlet wrapper classes	L2	CO3	2	Servlet API	
13.	Do we need to override service() method?	L3	CO3	2	Servlet API	
14.	How can we invoke another servlet in a different application?	L3	CO3	2	Servlet API	
15.	What is difference between GenericServlet and HttpServlet?	L2	CO3	2	Servlet API	
16.	Why HttpServlet class is declared abstract?	L2	CO3	2	Servlet API	
17.	Differentiate between ServletResponse sendRedirect() and RequestDispatcher forward() method	L2	CO3	2	Request Dispatcher	
18.	How do you handle exceptions and errors in a servlet	L3	CO3	2	Servlet API	
19.	How does Cookies work in Servlets?	L3	CO3	2	Cookies and Http Session	
20.	Describe handling file uploads in Servlets	L3	CO3	2	Working with Files	
21.	Show Setting up of the MIME type when serving a file from a servlet	L1	CO3	2	Working with Files	
22.	Compare traditional vs non-blocking I/O	L2	CO3	2	Non- blocking I/O in Servlets.	
23.	List the key interfaces and methods involved in Servlet non-blocking I/O	L1	CO3	2	Non- blocking I/O in Servlets.	
24.	Justify the use of Prepared Statement with an example	L3	CO3	2	Working with Databases through Servlets	
25.	Demonstrate session tracking and its techniques	L1	CO3	4	Session Management	
26.	How will you configure filter using deployment descriptor?	L1	CO3	4	Servlet API	
27.	What are the advantages and disadvantages of using cookies? Create an HTML file AddCookies.html, which contains four text boxes .When user enters the values in these text boxes and press the submit button, the values will be sent to AddCookies.java servlet and get saved in cookies	L2	CO3	4	Cookies and Http Session	
28.	Explain the different types of Java EE containers	L2	CO3	4	Java EE Architecture	
29.	Write the steps for creating servlet program with example program.	L2	CO3	4	Servlet API	
30.	Create a JDBC application to Insert User Registration details like RegistrationNo,Username,Password,Email into UserRegistration table. Also perform update and delete operations on specified rows.	L1	CO3	4	Working with Databases through Servlets	

	Explain in detail about handling HTTPRequest and HTTPResponse with examples.	L1	CO3	4	Handling Client Requests and Responses	
32.	Explain cookies in servlet with suitable example.	L2	CO3	4	Cookies and Http Session	
33.	Demonstrate about (i) Database programming using JDBC. b) Specific database actions.	L1	CO3	4	Working with Databases through Servlets	
34.	How does HTTP Servlet handle client requests?	L1	CO3	4	Handling Client Requests and Responses	
35.	Explain in detail about servlet parameters and reading initialization parameters.	L3	CO3	4	Handling Client Requests and Responses	
36.	Write a short note on how to obtain the object of RequestDispatcher	L2	CO3	4	Request Dispatcher	
37.	Develop a servlet program to perform CRUD (Create, Read, Update, Delete) operations with a database table	L3	CO3	6	Working with Databases through Servlets	
38.	Explain about sharing session, application between JSP pages in detail	L2	CO3	6	Session Management	
39.	Create an HTML form login.html that accepts the username and password and submits to login.java servlet. Verify credentials and on success, set the username in the HTTP Session and show a "Welcome <username>" message	L3	CO3	6	Servlet API	
40.	Explain how Cookies work? And where the Cookies are used?	L2	CO3	6	Cookies and Http Session	
41.	Write a servlet that uses Cookies to store the number of times a user has visited a servlet.	L2	CO3	6	Cookies and Http Session	
42.	Justify with Advantages and Disadvantages of Stateless nature. Explain how the session works	L4	CO3	6	Cookies and Http Session	
43.	Write a servlet program to insert Employee details like: empid,employee_name,address,date_of_birth in Employee table using JDBC	L3	CO3	6	Working with Databases through Servlets	
44.	Write a code to create a servlet for a login page. If the username and password are correct, then itsays message "Hello " else a message "login failed" will be displayed	L3	CO3	6	Servlet API	
45.	List four types of JDBC drivers and Explain any 2 of them in detail	L2	CO3	6	Working with Databases through Servlets	
46.	Enlist session management techniques and explain with example programs	L3	CO3	6	Session Management	
47.	Define Non-Blocking I/O? Explain WriteListener and ReadListener performing in	L2	CO3	8	Non- blocking I/O in Servlets.	

	Non-Blocking I/O					
48.	Explain the following: i) Setting a Cookie ii) Sending a Cookie to the Client iii) Removing Cookies	L4	CO3	12	Cookies and Http Session	
49.	Write a short note on HttpSession Interface. Also write the programming code for following: i) Creating a new Session ii) Getting a Pre-existing Session iii) Destroying a session	L4	CO3	12	Cookies and Http Session	
50.	Write a servlet program to create a session and display the following: i) Session ID ii) New or Old iii) Creation Time	L4	CO3	12	Session Management	
51.	Create a Login.html page and verify the user in LoginServlet.java servlet. If the user is valid then open the EmployeeForm.html otherwise display appropriate message in Login.html page	L4	CO3	16	Servlet API	
52.	Create an HTML form that accepts a username and password, and write a servlet to verify the credentials and set a session attribute upon successful login	L4	CO3	16	Servlet API	
53.	Develop a servlet program to count the number of visitors to a webpage using ServletContext	L4	CO3	16	Servlet API	
54.	Write a servlet program to count the number of visitors to a webpage using ServletContext	L4	CO3	16	Servlet API	
55.	Describe the basic flow for handling a non-blocking request and response and How would you handle a large file download using non-blocking I/O, write with all necessary methods.	L4	CO3	16	Request Dispatcher	